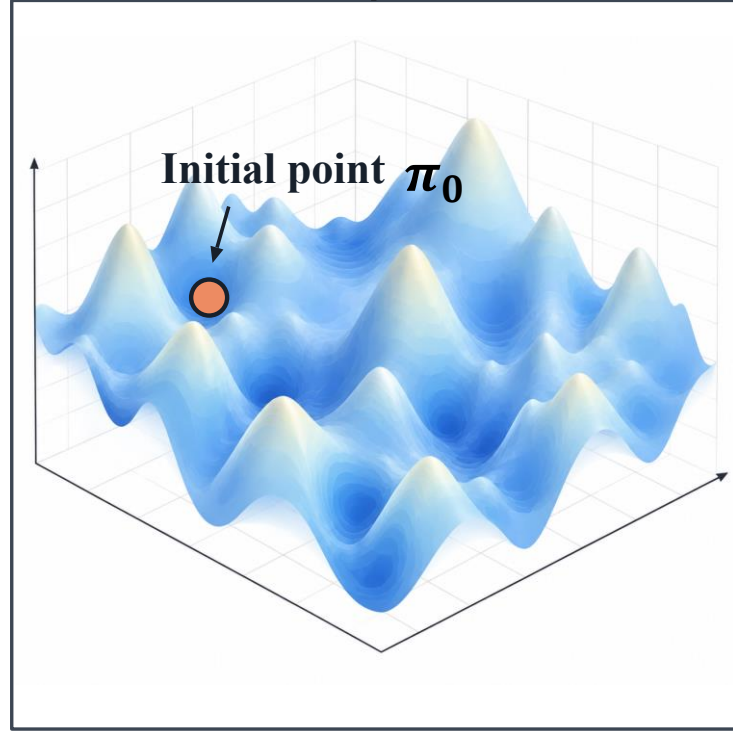
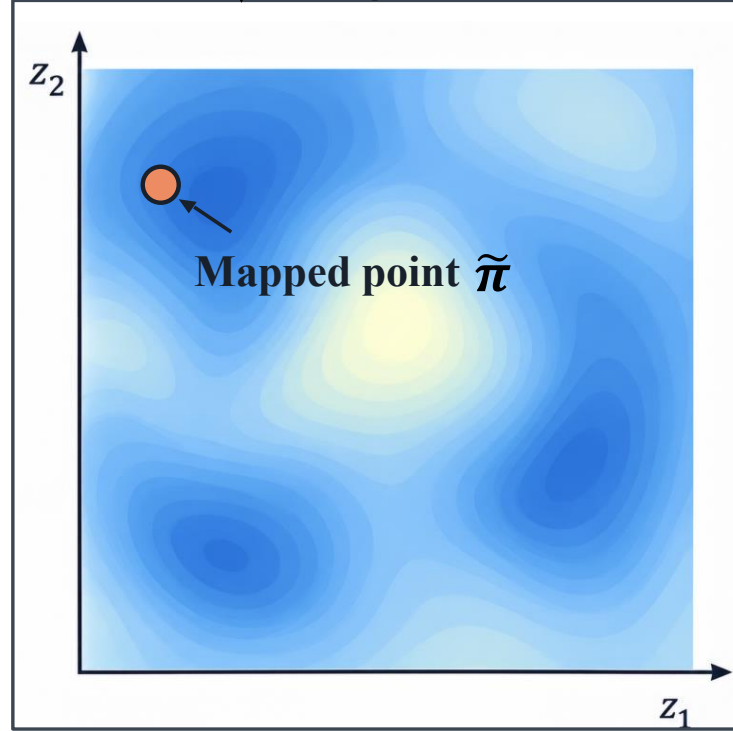


**Geometry-Guided Compression**

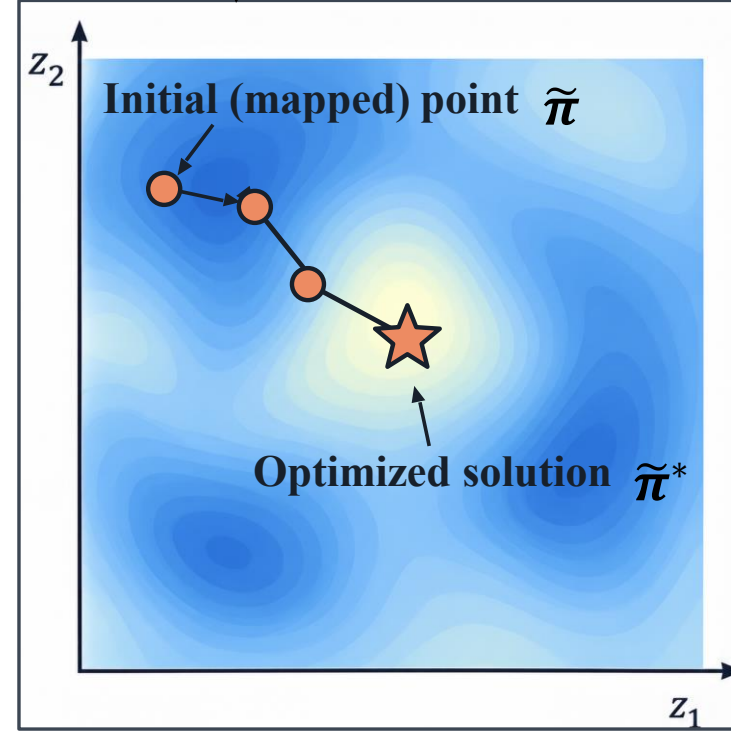
$$\tilde{\pi} = \Phi_S(\pi_0; M)$$

Original solution space  $\Omega(\mathbf{I})$ **Compact Space Optimization**

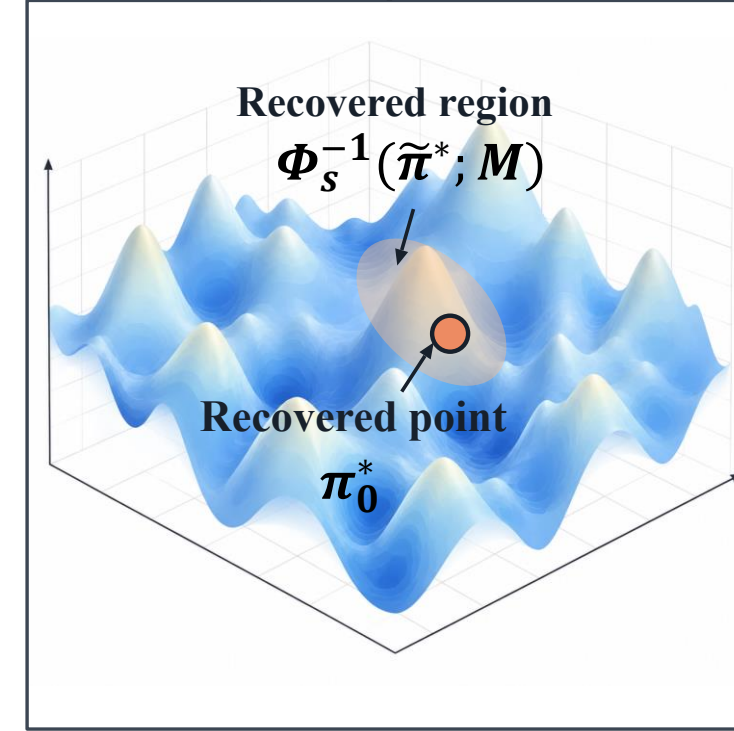
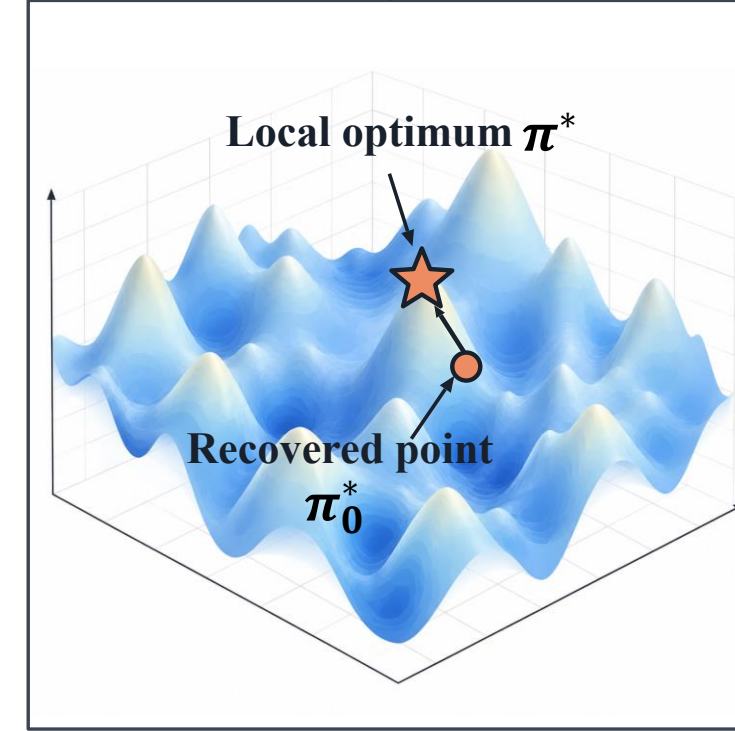
$$\tilde{\pi}_{k+1} = \arg \min_{\tilde{\pi}' \in N(\tilde{\pi}_k)} \hat{J}_B(\tilde{\pi}').$$

Compact solution space  $\Omega(\tilde{\mathbf{I}})$ **Initialization Recovery**

$$Pr(\pi_0^* = \pi | \tilde{\pi}^*, M) \propto \exp\left(\frac{g(\pi; \tilde{\pi}^*, M)}{\tau}\right)$$

Compact solution space  $\Omega(\tilde{\mathbf{I}})$ **Micro Refinement**

$$\pi_{k+1} = \arg \min_{\pi' \in N_I(\pi_k)} C_I(\pi')$$

Original solution space  $\Omega(\mathbf{I})$ Original solution space  $\Omega(\mathbf{I})$